

RayGlides 3W/4W EV Kit Cooling Controller - Manufacturing & Assembly Notes

This document provides production guidelines, board materials, soldering profiles, and wiring assembly specifications.

1. PCB Fabrication Specifications

- * **Material**: FR-4 TG150 (Glass transition temperature $\geq 150^{\circ}\text{C}$)
- * **Layer Count**: 2 Layers
- * **Board Thickness**: 1.6 mm
- * **Copper Weight (Outer)**: 2 oz/ft² (70 μm)
- * **Surface Finish**: HASL Lead-Free (RoHS compliant) or ENIG (Electroless Nickel Immersion Gold)
- * **Solder Mask Color**: Matte Green or Blue (LPI - Liquid Photoimageable)
- * **Silkscreen**: White (Lead-Free legibility)
- * **Minimum Drill Size**: 0.3 mm
- * **Minimum Trace/Space**: 0.15 mm (6 mils)

2. Component Assembly Instructions

- * **Solder Paste**: SAC305 (96.5% Sn, 3% Ag, 0.5% Cu) Lead-Free solder paste.
- * **SMD Reflow Profile**: Conforms to IPC/JEDEC J-STD-020E standard:
- * **Preheat Temp**: 150°C to 200°C for 60-120 seconds
- * **Time Above Liquidus** (217°C): 60-90 seconds
- * **Peak Reflow Temp**: 245°C to 250°C for 20-30 seconds
- * **Manual Assembly**:
- * Through-hole connectors J1, J2, J3, and temperature sensor U4 can be wave-soldered or hand-soldered at 350°C using lead-free wire (Sn96.5/Ag3.0/Cu0.5).
- * Ensure the exposed thermal pad of U1 (LM5017) is fully reflowed to the PCB thermal via pad to prevent regulator overheating.

3. Wiring and Harness Specifications

1. **Input Power Harness (J1)**:
 - * **Connector Shell**: Molex Micro-Fit 3.0 2-circuit housing (43645-0200)
 - * **Terminal Crimp Pin**: Molex male/female tin crimp terminal (43030-0007)
 - * **Wire Gauge**: **20 AWG** (Teflon or silicone insulation rated for 300V, 105°C operational temperature)

2. ****Fan Power Harness (J2)**:**

* ***Connector Shell*:** Molex Micro-Fit 3.0 2-circuit housing (43645-0200)

* ***Wire Gauge*:** ****20 AWG**** or ****22 AWG**** (Red: +12V, Black: switched GND return)

3. ****Sensor Cable Harness (J3)**:**

* ***Connector Shell*:** Molex Micro-Fit 3.0 3-circuit housing (43645-0300)

* ***Wire Gauge*:** ****24 AWG**** (shielded 3-conductor cable; shield must be connected to GND_SENS at the PCB side only to minimize EMI loop pickup)